ELLER COLLEGE OF MANAGEMENT
HSLopez School of Business Analytics

# ANALYTICS REPORT

**TO:** AMERICAN EXPRESS

**FROM:** BARRON BENBOW

**SUBJECT:** ANALYSIS OF TAIWANESE CREDIT CARD DATA

**DATE:** SEPTEMBER 23, 2025

## *Introduction*

As requested, I analyzed a dataset of close to 25,000 Taiwanese customers to determine how American Express can improve its marketing strategy. The purpose of this report is to evaluate whether customer spending patterns differ by gender and whether average credit card payments vary between April and September. To do so, I conducted two tests: the first compared male and female credit card bills, and the second compared the average monthly payments across the two selected months. The results showed that male customers have higher credit card bills, and that September has higher average credit card payments than April. Based on these findings, I recommend that American Express increase its advertising efforts during September and the surrounding months in order to maximize profit and better target customers.

## *Data Analysis*

There were two t-tests conducted. Section 1 compares the average bill amounts between males and females. Section 2 test compares the average payment amounts between April and September, as well as two tables showing a visualization on the data from the report.

<u>Section 1 – t-test Comparing Average Bill Amounts Between Males and Females</u>
*See the Technical Appendix for t-test output*

$H_0$: There is not a difference in average credit card bill amount between male and females.
$H_A$: There is a difference in average credit card bill amount between male and females.

Because the p-value = 9.65E-06 is less than the significance level of 0.05, we can reject the null hypothesis and can conclude there is a difference in average credit card bill amount between males, and females. Since we were able to show statistically significant difference, we can now look at the sample averages to see that males have higher average bill amounts than females, even for the population.

$$\left(1570.462 - 1445.296\right) \pm \sqrt{\frac{4919941.366}{9529} + \frac{4178620.342}{14760}}$$

To get a range of how much different male bills are than female bills, on average, we will use a confidence interval.

To get a range of how much different September are than April payments, on average, we will use a confidence interval [180.5829615, 69.74903846], since the interval does not include zero, we can say there is a difference.

Section 2 – t-test Comparing Average Payment Amounts Between April and September
*See the Technical Appendix for t-test output*

$H_0$: There is not a difference in average credit card payment amounts between April and September

$H_A$: There is a difference in average credit card payment amounts between April and September

Because the p-value = 0.000906076767525576 is less than the significance level of 0.05, we can reject the null hypothesis and can conclude there is a difference in average credit card bill amount between males, and females. Since we were able to show statistically significant difference, we can now look at the sample averages to see that males have higher average bill amounts than females, even for the population.

$$15.15281498 \pm 1.9600626\left(\frac{771.6150933}{\sqrt{2489}}\right)$$

To get a range of how much different April are than September payments, on average, we will use a confidence interval. The resulting interval [24.10255231,6.203077651], since the interval does not include zero, we can say there is a difference.

Section 3 – Recommendations from Testing Results

I recommend American Express to have more commercials during September and the months before and after to maximize profit. I would also recommend finding a way to target men in a more centralized way.

***Conclusion***

In conclusion, both tests offered great benefits to expand profit. From conducting these tests I have found that males credit card bill's are higher and that September has higher average credit card payments than April. From my findings I would recommend more advertising for American Express targeting a broader male audience. For future analysis we could combine both tests to find what month gets billed the highest

for male and females. I hope this information is beneficial to the company, please feel free to reach out to me using my contact information below with any questions, concerns, or more analysis on the data.

*Contact Information*

American Express

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***Technical Apendix***

*Figure 1 – t-test Comparing Average Bill Amounts Between Males and Females*

| t-Test: Two-Sample Assuming Unequal Variances | | |
|---|---|---|
| | | |
| | *Male Bills* | *Female Bills* |
| Mean | 1570.462231 | 1445.296056 |
| Variance | 4919941.366 | 4178620.342 |
| Observations | 9529 | 14760 |
| Hypothesized Mean Difference | 0 | |
| df | 19129 | |
| t Stat | 4.42690642 | |
| P(T<=t) one-tail | 4.80616E-06 | |
| t Critical one-tail | 1.644933288 | |
| P(T<=t) two-tail | 9.61232E-06 | |
| t Critical two-tail | 1.960088007 | |
| Barron Benbow | | |

*Figure 2 – t-test Comparing Average Payment Amounts Between April and September*

| t-Test: Paired Two Sample for Means | | |
|---|---|---|
| | | |
| | *September Payment* | *April Payment* |
| Mean | 185.8194757 | 170.6666607 |
| Variance | 272014.5958 | 345387.4292 |
| Observations | 24289 | 24289 |
| Pearson Correlation | 0.181078558 | |
| Hypothesized Mean Difference | 0 | |
| df | 24288 | |
| t Stat | 3.318585314 | |
| P(T<=t) one-tail | 0.000453038 | |
| t Critical one-tail | 1.644916367 | |
| P(T<=t) two-tail | 0.000906077 | |
| t Critical two-tail | 1.960061662 | |
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